



NATIONAL UNIVERSITY OF MODERN LANGUAGES

Department of Software Engineering

BSSE 5 (Afternoon) – Lab Final–Fall 2024

Paper: CN Lab

Time Allowed: 3 Hours

Total Marks: 50

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Report Submission

Once you conclude the configuration paste the figure(s) in the desired Section, further instructions are in the sections.

Section 1: LANs

[20 Marks]

Section 1.1: Switch Running-configuration [3 Marks]

Paste the running-configurations Figure under each section. Show the IP address configured for each VLAN as per Figure 1.

NOTE: Each Switch should have MOTD and Name as defined in Q1 bullet 4.

A. Switch 1 Running Config

```
zulqarnain-sp21379#show running-config
Building configuration...

Current configuration : 2285 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname zulqarnain-sp21379
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
switchport mode trunk
!
interface FastEthernet0/2
switchport mode trunk
!
interface FastEthernet0/3
switchport mode trunk
!
interface FastEthernet0/4
switchport mode access
!
interface FastEthernet0/5
switchport mode access
!
interface FastEthernet0/6
switchport mode access
!
interface FastEthernet0/7
switchport access vlan 66
switchport mode access
!
```

B. Switch 2 Running Config

```
zulqarnain-sp21379#show r
zulqarnain-sp21379#show running-config
Building configuration...

Current configuration : 2254 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname zulqarnain-sp21379
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
  switchport mode trunk
!
interface FastEthernet0/2
  switchport mode trunk
!
interface FastEthernet0/3
  switchport mode access
!
interface FastEthernet0/4
  switchport mode access
!
interface FastEthernet0/5
  switchport mode access
!
interface FastEthernet0/6
  switchport access vlan 66
  switchport mode access
!
interface FastEthernet0/7
  switchport access vlan 66
--More--
```

C. Switch 3 Running Config

```

zulqarnain-sp21379#show run
zulqarnain-sp21379#show running-config
Building configuration...

Current configuration : 2286 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname zulqarnain-sp21379
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 switchport mode trunk
!
interface FastEthernet0/2
 switchport mode trunk
!
interface FastEthernet0/3
 switchport mode access
!
interface FastEthernet0/4
 switchport mode access
!
interface FastEthernet0/5
 switchport mode access
!
interface FastEthernet0/6
 switchport mode access
!

```

Section 1.2: Switch VLAN brief - Ports configuration [3 Marks]

Paste the VLANs ports distribution under each section.

A. Switch 1 VLAN Ports

```

zulqarnain-sp21379#show vlan b
zulqarnain-sp21379#show vlan brief

```

VLAN	Name	Status	Ports
1	default	active	Fa0/4, Fa0/5, Fa0/6, Gig0/1 Gig0/2
66	students	active	Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15
99	faculty	active	Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

B. Switch 2 VLAN Ports

```

zulqarnain-sp21379#show vlan b
zulqarnain-sp21379#show vlan brief

```

VLAN	Name	Status	Ports
1	default	active	Fa0/3, Fa0/4, Fa0/5, Gig0/1 Gig0/2
66	students	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15
99	faculty	active	Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	
zulqarnain-sp21379#			
zulqarnain-sp21379#			
zulqarnain-sp21379#			

C. Switch 3 VLAN Ports

```

zulqarnain-sp21379#show vlan brief

```

VLAN	Name	Status	Ports
1	default	active	Fa0/3, Fa0/4, Fa0/5, Fa0/6 Gig0/1, Gig0/2
66	students	active	Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15
99	faculty	active	Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	
zulqarnain-sp21379#			
zulqarnain-sp21379#			
zulqarnain-sp21379#			
zulqarnain-sp21379#			

Section 1.3: Switch Root information [3 Marks]

Paste the spanning tree protocol information showing that Switch 1 is configured as a ROOT Switch.

A. Switch 1 Root Switch

```

zulqarnain-sp21379#show spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
    Root ID      Priority    32769
                Address      0002.1708.D87D
                Cost         19
                Port         2(FastEthernet0/2)
                Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

    Bridge ID    Priority    32769 (priority 32768 sys-id-ext 1)
                Address      00D0.97E5.6A6D
                Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
                Aging Time   20

Interface        Role Sts Cost      Prio.Nbr Type
-----
Fa0/1             Desg FWD 19        128.1    P2p
Fa0/3             Altn BLK 19        128.3    P2p
Fa0/2             Root FWD 19        128.2    P2p

```

```

VLAN0066
  Spanning tree enabled protocol ieee
    Root ID      Priority    32834
                Address      0002.1708.D87D
                Cost         19
                Port         2(FastEthernet0/2)
                Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

    Bridge ID    Priority    32834 (priority 32768 sys-id-ext 66)
                Address      00D0.97E5.6A6D
                Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
                Aging Time   20

```

```

Interface        Role Sts Cost      Prio.Nbr Type
-----
Fa0/7             Desg FWD 19        128.7    P2p
Fa0/1             Desg FWD 19        128.1    P2p
Fa0/8             Desg FWD 19        128.8    P2p
Fa0/3             Altn BLK 19        128.3    P2p
Fa0/2             Root FWD 19        128.2    P2p

```

```

VLAN0099

```

Section 1.4: DHCP Server [2 Marks]

Paste the DHCP configuration figures of each DHCP under the desired heading.

A. VLAN666-DHCP

DHCP VLAN66

PhysicalConfigServicesDesktopProgrammingAttributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

DHCP

InterfaceFastEthernet0ServiceOnOff

Pool NameserverPool

Default Gateway192.66.79.1

DNS Server0.0.0.0

Start IP Address :192667910

Subnet Mask:255255255224

Maximum Number of Users :22

TFTP Server:0.0.0.0

WLC Address:0.0.0.0

AddSaveRemove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	192.66.79.1	0.0.0.0	192.66.79.10	255.255.2...	22	0.0.0.0	0.0.0.0

B. VLAN999-DHCP

SERVICES

- HTTP
- DHCP**
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP

DHCP

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: serverPool

Default Gateway: 192.99.79.1

DNS Server: 0.0.0.0

Start IP Address: 192.99.79.10

Subnet Mask: 255.255.255.224

Maximum Number of Users: 22

TFTP Server: 0.0.0.0

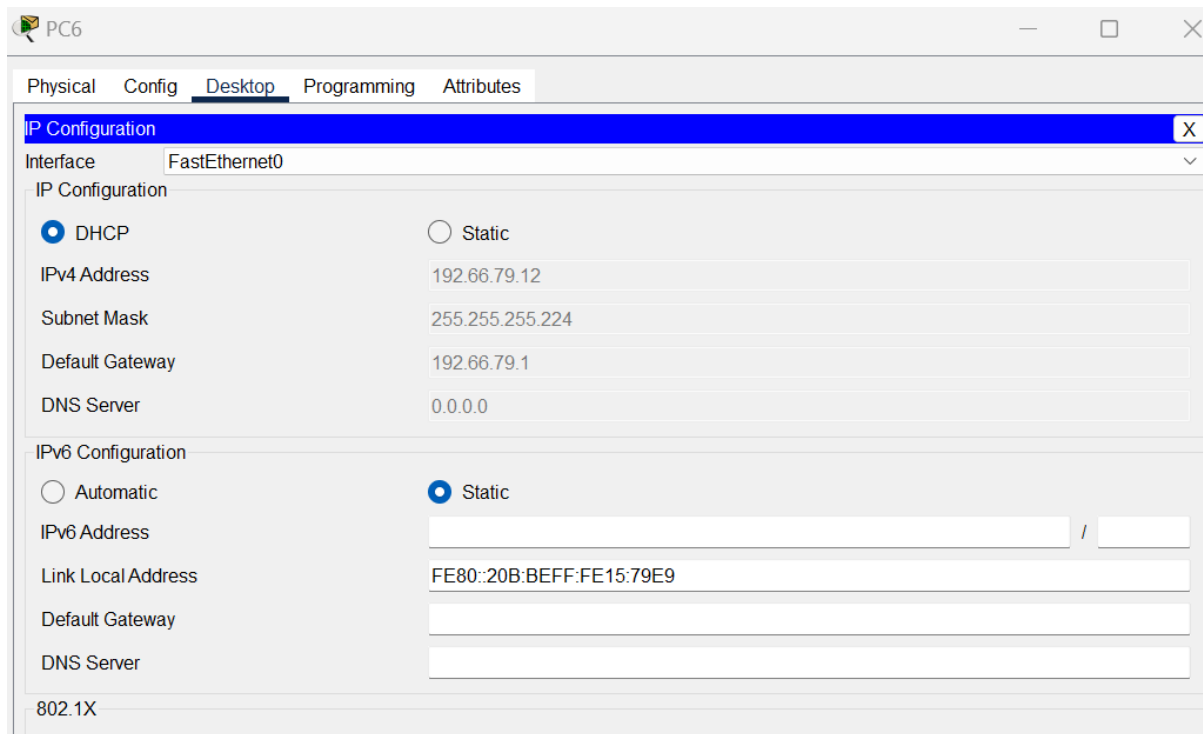
WLC Address: 0.0.0.0

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	192.99.79.1	0.0.0.0	192.99.79.10	255.255.255.224	22	0.0.0.0	0.0.0.0

Section 1.5: DHCP working [2 Marks]

Show a PC has received an IP address from its corresponding DHCP servers.

A. PC of VLAN666



PC6

Physical Config **Desktop** Programming Attributes

IP Configuration [X]

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static

IPv4 Address 192.66.79.12

Subnet Mask 255.255.255.224

Default Gateway 192.66.79.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address

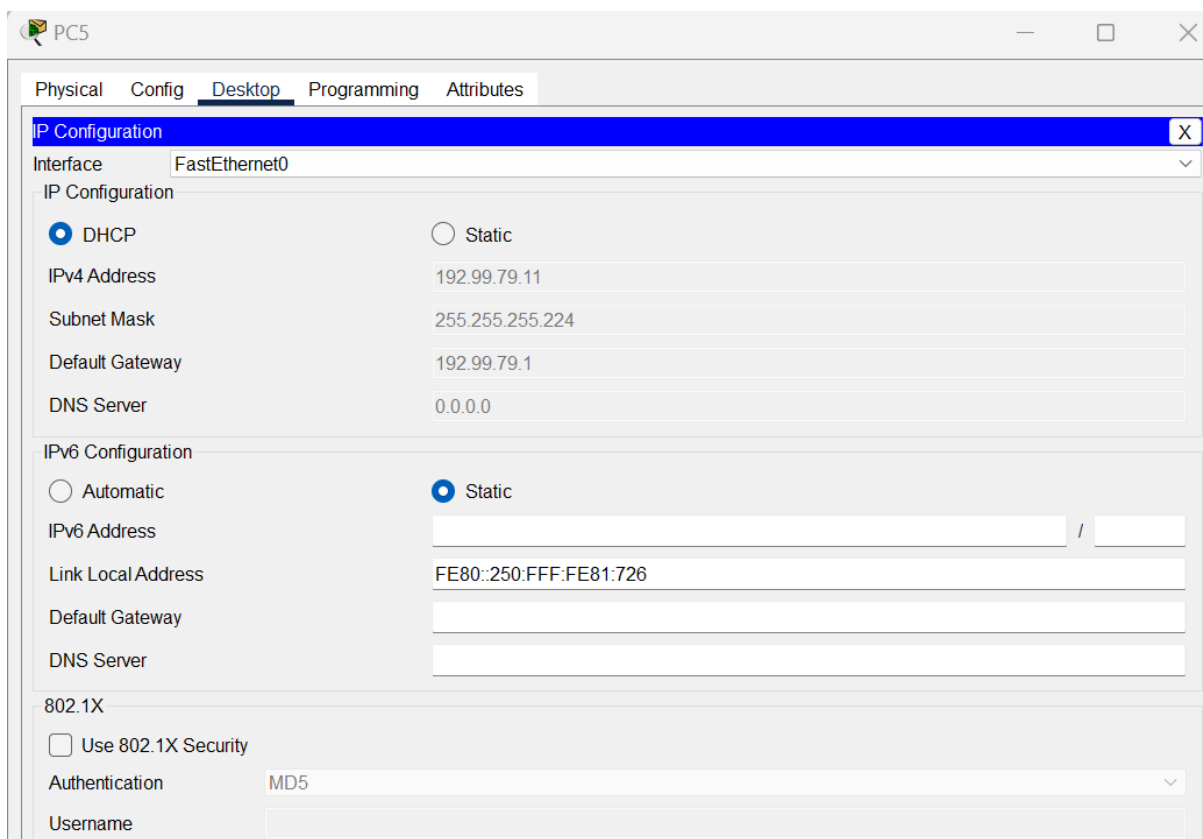
Link Local Address FE80::20B:BEFF:FE15:79E9

Default Gateway

DNS Server

802.1X

B. PC of VLAN999



PC5

Physical Config **Desktop** Programming Attributes

IP Configuration [X]

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static

IPv4 Address 192.99.79.11

Subnet Mask 255.255.255.224

Default Gateway 192.99.79.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address

Link Local Address FE80::250:FFF:FE81:726

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

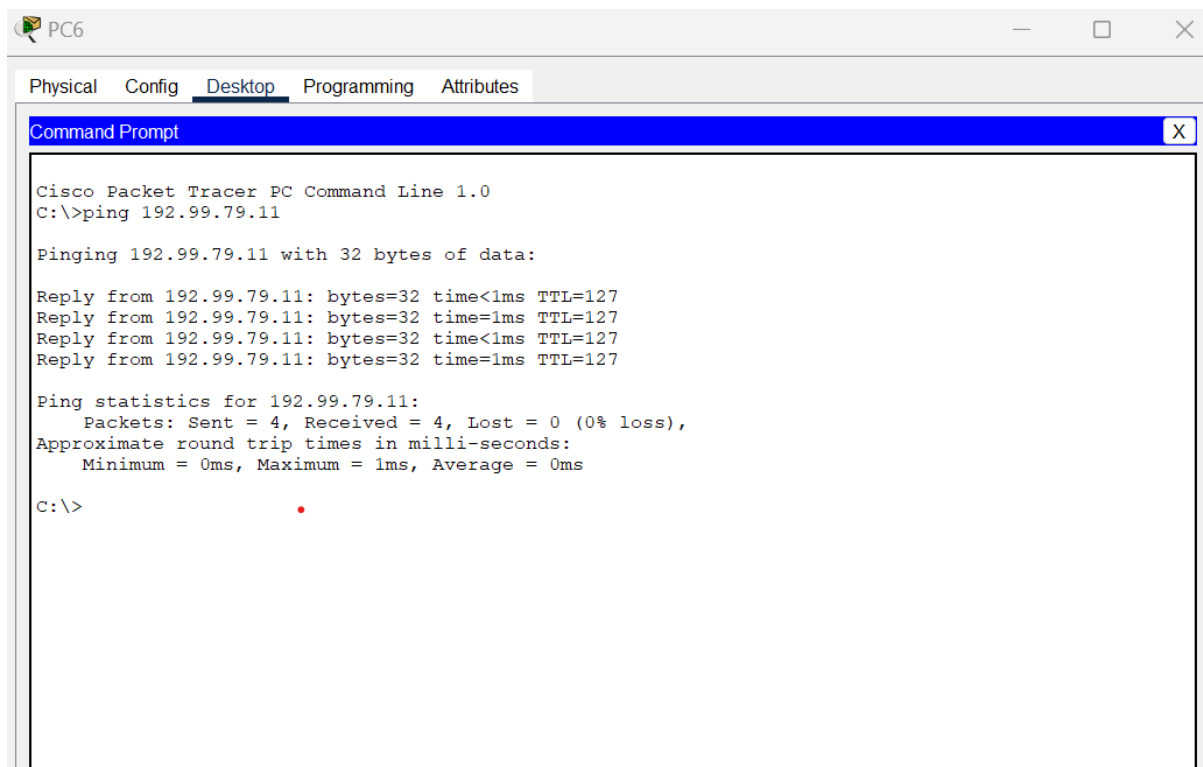
Authentication MD5

Username

Section 1.6: Inter-VLAN-Routing – Pinging VLANs [3 Marks]

Show a PC belonging to VLAN 666 PING to PC of VLAN999.

A. VLAN666 PC PING VLAN999



The screenshot shows a Cisco Packet Tracer PC Command Line window for PC6. The 'Desktop' tab is selected. The command prompt displays the output of a ping command to 192.99.79.11. The output shows four successful replies with 32 bytes of data, a time of less than 1ms, and a TTL of 127. The ping statistics show 4 packets sent, 4 received, and 0% loss, with round trip times of 0ms.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.99.79.11

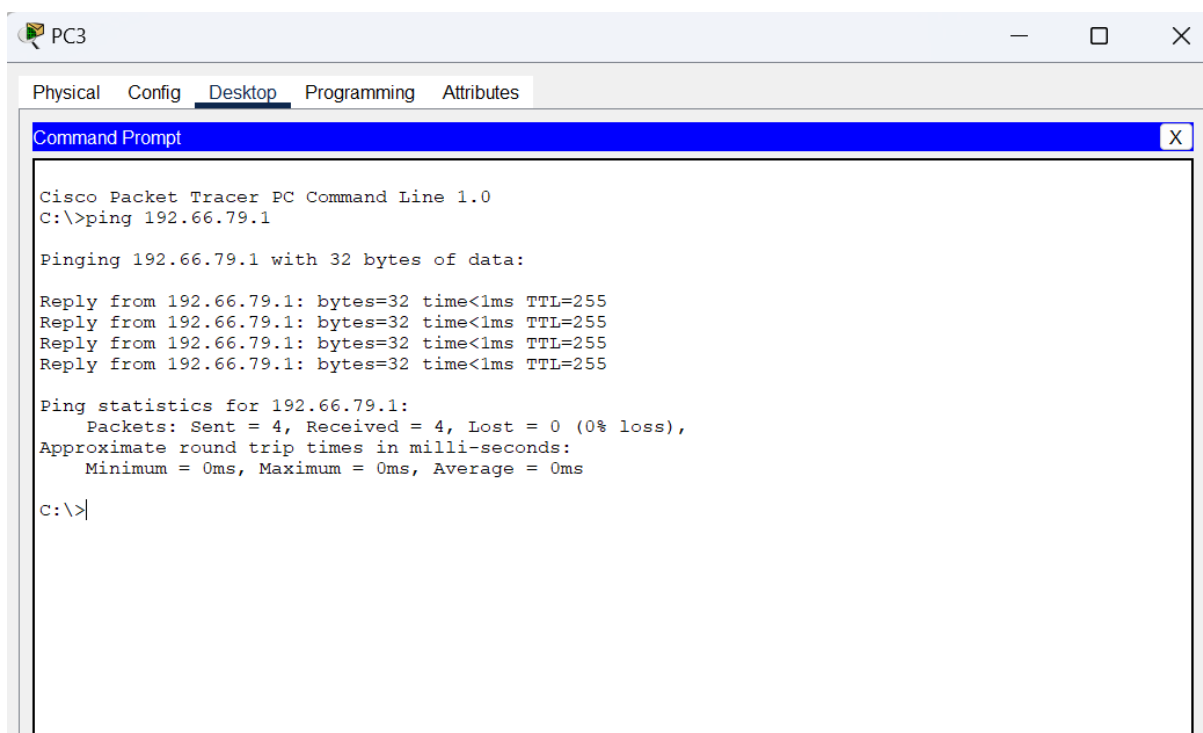
Pinging 192.99.79.11 with 32 bytes of data:

Reply from 192.99.79.11: bytes=32 time<1ms TTL=127
Reply from 192.99.79.11: bytes=32 time=1ms TTL=127
Reply from 192.99.79.11: bytes=32 time<1ms TTL=127
Reply from 192.99.79.11: bytes=32 time=1ms TTL=127

Ping statistics for 192.99.79.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

B. VLAN666 PC PING its Gaetway



The screenshot shows a Cisco Packet Tracer PC Command Line window for PC3. The 'Desktop' tab is selected. The command prompt displays the output of a ping command to 192.66.79.1. The output shows four successful replies with 32 bytes of data, a time of less than 1ms, and a TTL of 255. The ping statistics show 4 packets sent, 4 received, and 0% loss, with round trip times of 0ms.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.66.79.1

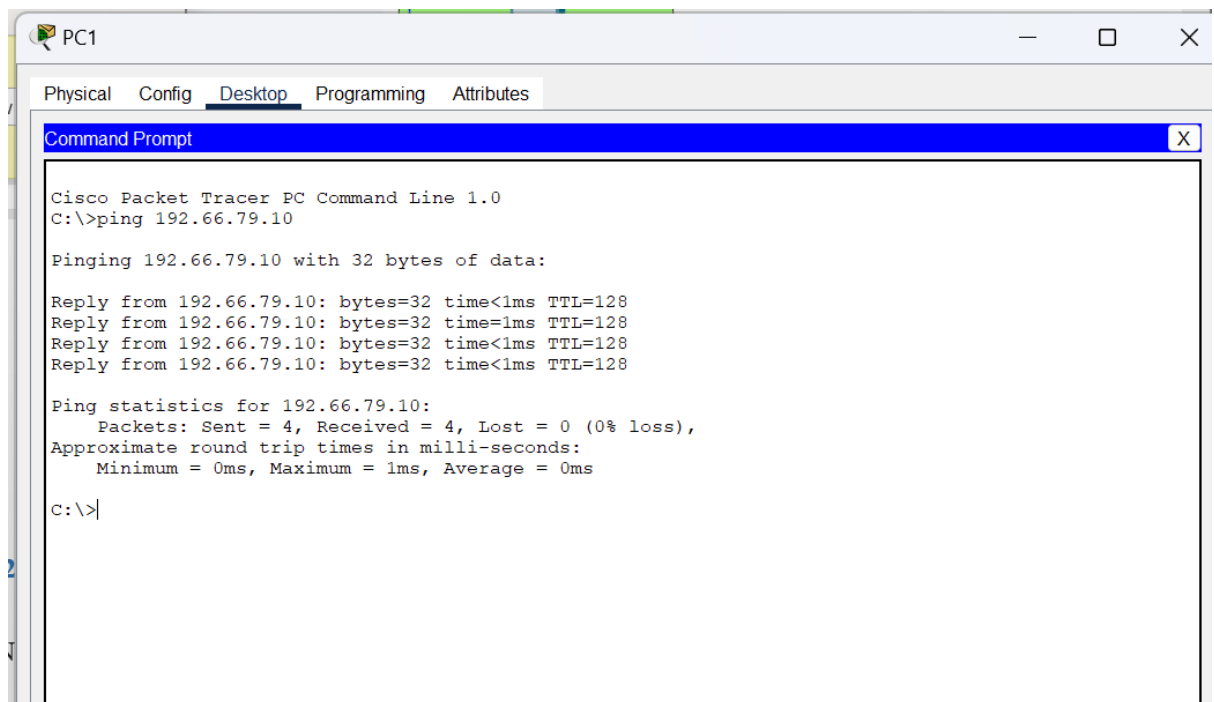
Pinging 192.66.79.1 with 32 bytes of data:

Reply from 192.66.79.1: bytes=32 time<1ms TTL=255
Reply from 192.66.79.1: bytes=32 time<1ms TTL=255
Reply from 192.66.79.1: bytes=32 time<1ms TTL=255
Reply from 192.66.79.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.66.79.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

C. VLAN999 PC PING VLAN666



D. VLAN999 PC PING its Gaetway

```
C:\>ping 192.99.79.1

Pinging 192.99.79.1 with 32 bytes of data:

Reply from 192.99.79.1: bytes=32 time<1ms TTL=255
Reply from 192.99.79.1: bytes=32 time=1ms TTL=255
Reply from 192.99.79.1: bytes=32 time<1ms TTL=255
Reply from 192.99.79.1: bytes=32 time=1ms TTL=255

Ping statistics for 192.99.79.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Section 1.7: Router on the stick – Lahore Router – Routing Table [2 Marks]

Show the IP Route of Lahore Router highlighting the 802.1q protocol for all VLANs. Provide description of the interface as well, for convenience.

A. Lahore Router – Routing Table

```

zulqarnain-21379#shw
zulqarnain-21379#sho
zulqarnain-21379#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

C    110.0.0.0/8 is directly connected, Serial0/0
    192.66.79.0/27 is subnetted, 1 subnets
C    192.66.79.0 is directly connected, FastEthernet0/0.66
    192.99.79.0/27 is subnetted, 1 subnets
C    192.99.79.0 is directly connected, FastEthernet0/0.99
R    192.168.2.0/24 [120/1] via 110.10.1.2, 00:00:05, Serial0/0
R    192.168.3.0/24 [120/1] via 110.10.1.2, 00:00:05, Serial0/0
    192.168.79.0/27 is subnetted, 1 subnets
C    192.168.79.0 is directly connected, FastEthernet0/0.1

zulqarnain-21379#

```

Section 1.8: Router on the stick – Lahore Router – Routing Table [2 Marks]

Show the IP Route of Lahore Router highlighting the 802.1q protocol for all VLANs. Provide description of each interface as well, for convenience.

A. Lahore Router – Routing Table

```

zulqarnain-21379#shw
zulqarnain-21379#sho
zulqarnain-21379#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

C    110.0.0.0/8 is directly connected, Serial0/0
    192.66.79.0/27 is subnetted, 1 subnets
C    192.66.79.0 is directly connected, FastEthernet0/0.66
    192.99.79.0/27 is subnetted, 1 subnets
C    192.99.79.0 is directly connected, FastEthernet0/0.99
R    192.168.2.0/24 [120/1] via 110.10.1.2, 00:00:05, Serial0/0
R    192.168.3.0/24 [120/1] via 110.10.1.2, 00:00:05, Serial0/0
    192.168.79.0/27 is subnetted, 1 subnets
C    192.168.79.0 is directly connected, FastEthernet0/0.1

zulqarnain-21379#

```

Section 2: WAN**[10 Marks]****Section 2.1: Routing Running Configuration of ALL Routers [3 Marks]**

Once the Routers are configured, paste the Figure of each designated Router's running-configuration under each section.

NOTE: Each Router should have MOTD and Name as defined in Q1 bullet 4.

A. Router Lahore Running Config

```
zulqarnain-21379#show running-config
Building configuration...

Current configuration : 1007 bytes
!
version 12.2
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname zulqarnain-21379
!
!
!
!
!
!
!
ip cef
no ipv6 cef
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
interface FastEthernet0/0
  no ip address
  duplex auto
```

```

no ip address
duplex auto
speed auto
!
interface FastEthernet0/0.1
encapsulation dot1Q 1 native
ip address 192.168.79.1 255.255.255.224
!
interface FastEthernet0/0.66
encapsulation dot1Q 66
ip address 192.66.79.1 255.255.255.224
!
interface FastEthernet0/0.99
encapsulation dot1Q 99
ip address 192.99.79.1 255.255.255.224
!
interface Serial0/0
ip address 110.10.1.1 255.0.0.0
!
interface Serial0/1
no ip address
clock rate 2000000
!
interface FastEthernet1/0
no ip address
duplex auto
speed auto
!
router rip
version 2
network 110.0.0.0
network 192.66.79.0
network 192.99.79.0
network 192.168.1.0
network 192.168.79.0
!
ip classless
!
ip flow-export version 9
!
!
!
```

B. Router PTCL Running Config

```
zulgarnian-sp21379#show running-config
Building configuration...

Current configuration : 673 bytes
!
version 12.2
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname zulgarnian-sp21379
!
!
!
!
!
!
!
ip cef
no ipv6 cef
!
!
!
!
!
!
!
!
!
!
interface FastEthernet0/0
 ip address 192.168.2.1 255.255.255.0
 duplex auto
```

```

interface FastEthernet0/0
 ip address 192.168.2.1 255.255.255.0
 duplex auto
 speed auto
!
interface Serial0/0
 ip address 110.10.1.2 255.0.0.0
 clock rate 2000000
!
interface Serial0/1
 ip address 192.168.3.1 255.255.255.0
 clock rate 2000000
!
router rip
 version 2
 network 110.0.0.0
 network 192.168.2.0
 network 192.168.3.0
!
ip classless
!
ip flow-export version 9
!
!
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
 login
!
!
!
end

```

C. Router HQISB Running Config

[illegible]

```

!
interface FastEthernet0/0
 ip address 192.168.79.1 255.255.255.0
 duplex auto
 speed auto
!
interface Serial0/0
 no ip address
 clock rate 2000000
!
interface Serial0/1
 ip address 192.168.3.2 255.255.255.0
!
router rip
 version 2
 network 192.168.3.0
 network 192.168.79.0
!
ip classless
!
ip flow-export version 9
!
!
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
 login
!
!
!
end

```

```

zulqarnain-sp21379#
zulqarnain-sp21379#

```

Section 2.2: Routers IP Route [3 Marks]

Once the Routers are configured, paste the Figures of each designated Router's IP Route under each section.

A. Router Lahore IP Route

```

zulqarnain-21379#show ip ro
zulqarnain-21379#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

C    110.0.0.0/8 is directly connected, Serial0/0
    192.66.79.0/27 is subnetted, 1 subnets
C    192.66.79.0 is directly connected, FastEthernet0/0.66
    192.99.79.0/27 is subnetted, 1 subnets
C    192.99.79.0 is directly connected, FastEthernet0/0.99
R    192.168.2.0/24 [120/1] via 110.10.1.2, 00:00:12, Serial0/0
R    192.168.3.0/24 [120/1] via 110.10.1.2, 00:00:12, Serial0/0
    192.168.79.0/27 is subnetted, 1 subnets
C    192.168.79.0 is directly connected, FastEthernet0/0.1

zulqarnain-21379#
zulqarnain-21379#

```

B. Router PTCL IP Route

```

zulqarnian-sp21379#show ip ro
zulqarnian-sp21379#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

C    110.0.0.0/8 is directly connected, Serial0/0
R    192.66.79.0/24 [120/1] via 110.10.1.1, 00:00:02, Serial0/0
R    192.99.79.0/24 [120/1] via 110.10.1.1, 00:00:02, Serial0/0
C    192.168.2.0/24 is directly connected, FastEthernet0/0
C    192.168.3.0/24 is directly connected, Serial0/1
R    192.168.79.0/24 [120/1] via 110.10.1.1, 00:00:02, Serial0/0
        [120/1] via 192.168.3.2, 00:00:28, Serial0/1

zulqarnian-sp21379#

```

C. Router HQISB IP Route

```

zulqarnain-sp21379#show ip ro
zulqarnain-sp21379#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

R    110.0.0.0/8 [120/1] via 192.168.3.1, 00:00:19, Serial0/1
R    192.66.79.0/24 [120/2] via 192.168.3.1, 00:00:19, Serial0/1
R    192.99.79.0/24 [120/2] via 192.168.3.1, 00:00:19, Serial0/1
R    192.168.2.0/24 [120/1] via 192.168.3.1, 00:00:19, Serial0/1
C    192.168.3.0/24 is directly connected, Serial0/1
C    192.168.79.0/24 is directly connected, FastEthernet0/0

zulqarnain-sp21379#

```

Section 2.3: DNS Configuration [4 Marks]

Once the DNS is configured, paste the Figures showing the entries of the switch and routers against their designated IP addresses, under the section.

A. DNS Configuration

DNS Server

Physical Config **Services** Desktop Programming Attributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

DNS

DNS Service

☐ On

☒ Off

Resource Records

NameType

A Record

Address

Add

Save

Remove

No.	Name	Type	Detail
0	ptcl	A Record	192.168.2.1

BSSE

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Section 2: Testing Network Connectivity [16 Marks]

Section 3.1: Testing the connectivity from PC0 [5 Marks]

Paste the Figure showing Ping information of the Following

A. PC0 to PC4 test

```
Pinging 192.99.79.10 with 32 bytes of data:

Reply from 192.99.79.10: bytes=32 time=2ms TTL=125
Reply from 192.99.79.10: bytes=32 time=3ms TTL=125
Reply from 192.99.79.10: bytes=32 time=2ms TTL=125
Reply from 192.99.79.10: bytes=32 time=11ms TTL=125

Ping statistics for 192.99.79.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 11ms, Average = 4ms
```

```
C:\>
```

B. PC0 to PC1 test

```
usage: ping [-n count] [-v tos] [-t] target

C:\>ping 192.66.79.11

Pinging 192.66.79.11 with 32 bytes of data:

Reply from 192.66.79.11: bytes=32 time=2ms TTL=125
Reply from 192.66.79.11: bytes=32 time=3ms TTL=125
Reply from 192.66.79.11: bytes=32 time=2ms TTL=125
Reply from 192.66.79.11: bytes=32 time=2ms TTL=125

Ping statistics for 192.66.79.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms
```

```
C:\>
```

Section 3.2: Testing the connectivity from PC1 [5 Marks]

Paste the Figure showing Ping information of the Following PCs

A. PC1 to PC4 test

```
C:\>ping 192.99.79.10

Pinging 192.99.79.10 with 32 bytes of data:

Reply from 192.99.79.10: bytes=32 time=1ms TTL=127
Reply from 192.99.79.10: bytes=32 time<1ms TTL=127
Reply from 192.99.79.10: bytes=32 time<1ms TTL=127
Reply from 192.99.79.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.99.79.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

B. PC1 to VLAN999 DHCP Server test

```
C:\>ping 192.99.79.5

Pinging 192.99.79.5 with 32 bytes of data:

Reply from 192.99.79.5: bytes=32 time<1ms TTL=127
Reply from 192.99.79.5: bytes=32 time<1ms TTL=127
Reply from 192.99.79.5: bytes=32 time<1ms TTL=127
Reply from 192.99.79.5: bytes=32 time<1ms TTL=127

Ping statistics for 192.99.79.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
C:\>
~.\>
```

C. PC1 to DNS Server

```
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time=1ms TTL=126
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126
Reply from 192.168.2.2: bytes=32 time=27ms TTL=126
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 27ms, Average = 7ms

C:\>
```

Section 3.3: Telnet to the Switch [6 Marks]

Once the Passwords are set Paste the Figure showing Telnet to of the designated Switches under the following headings.

A. PC5 to Switch 1

```
Reply from 192.99.79.1: bytes=32 time=1ms TTL=255

Ping statistics for 192.99.79.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>telnet 192.66.79.2
Trying 192.66.79.2 ...Open

User Access Verification

Password:
zulqarnain-sp21379>
zulqarnain-sp21379>
zulqarnain-sp21379>
```

B. PC1 to Switch 2

```
C:\>  
C:\>  
C:\>telnet 192.99.79.3  
Trying 192.99.79.3 ...Open  
  
User Access Verification  
  
Password:  
zulqarnain-sp21379>  
zulqarnain-sp21379>
```

C. PC3 to Switch 3

```
C:\>  
C:\>  
C:\>telnet 192.99.79.3  
Trying 192.99.79.3 ...Open  
  
User Access Verification  
  
Password:  
zulqarnain-sp21379>  
zulqarnain-sp21379>
```

*****THE END*****